

Siddharth Gautam

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SUMMARY

Electronics and Communication Engineering undergraduate at NSUT Delhi with hands-on experience in RTL design, digital verification, and FPGA prototyping. Proficient in Verilog/SystemVerilog with demonstrated projects spanning processor microarchitecture, communication protocols, and embedded systems. Seeking opportunities in VLSI design, hardware verification, or embedded systems engineering.

EDUCATION

Netaji Subhas University of Technology (NSUT), Delhi

Delhi, India

B.Tech – Electronics and Communication Engineering

2023 – 2027

EXPERIENCE

Research Intern

May 2025 – Jul 2025

NSUT, Delhi

Delhi, India

- Conducted computational modeling of $\text{Cs}_3\text{Bi}_2\text{I}_9$ perovskite as a non-toxic alternative for sustainable thin-film photovoltaic applications using SCAPS-1D simulation software.
- Optimized device architectures by systematically tuning absorber layer thickness (200–600 nm) and carrier concentrations, achieving a simulated power conversion efficiency improvement of over 15%.

PROJECTS

RISC-V 5-Stage Pipelined Processor | *Verilog, Xilinx Vivado*

🔗 Source Code

- Designed and implemented a 32-bit RISC-V (RV32I) CPU in **Verilog** with a 5-stage pipeline (IF/ID/EX/MEM/WB), full hazard detection unit, and data-forwarding paths to eliminate pipeline stalls.
- Achieved a post-synthesis clock frequency of **96 MHz** on Xilinx Artix-7 FPGA by reducing critical-path delay; validated correctness across 50+ test programs via GTKWave simulation.

SPI Protocol Controller | *System Verilog, GTKWave*

🔗 Source Code

- Developed a parameterized **SPI Master-Slave** controller in SystemVerilog supporting all four SPI modes (CPOL/CPHA), configurable clock dividers, and 8-to-32-bit frame widths.
- Achieved **100% functional coverage** using SystemVerilog Assertions (SVA) and a constrained-random UVM-style testbench environment, catching 3 RTL bugs before tape-out simulation.

UART Transmitter & Receiver | *Verilog, FPGA*

🔗 Source Code

- Implemented a fully synthesizable UART core with programmable baud rate generation (9600–115200), configurable parity (odd/even/none), and **FWFT FIFO** buffering for burst traffic handling.
- Reduced LUT utilization by **22%** and Flip-Flop usage by **18%** versus a baseline implementation through RTL micro-optimizations, enabling deployment on resource-constrained Spartan-7 devices.

DIY Prosthetic Hand Control System | *Embedded C++, Arduino*

🔗 Source Code

- Designed a real-time embedded control system using **Muscle Bio-Amp patchy** sensors and Arduino Nano to acquire sEMG signals and translate muscle activation patterns into servo motor movements.
- Developed adaptive thresholding algorithms in **Embedded C++** achieving end-to-end actuation latency under **50 ms**, with a modular software architecture enabling easy addition of new gesture profiles.

TECHNICAL SKILLS

HDL / Languages: Verilog, SystemVerilog, C/C++, Python (scripting & automation), Shell Scripting, Assembly (RISC-V)

EDA Tools: Xilinx Vivado, Cadence Virtuoso, GTKWave, ModelSim

Verification: SystemVerilog Assertions (SVA), Constrained-Random Simulation, Functional Coverage

Hardware & Protocols: RTL Design, Digital Logic Synthesis, FPGA Prototyping, SPI, UART, I2C

Core Subjects: Digital System Design, Computer Architecture, VLSI Design, Embedded Systems

Other: Git/GitHub, Linux/Bash, Technical Documentation